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A Novel Mobile Application to Empower Visually Impaired Individuals to Perform Independent Day-to-Day Activities

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***Abstract*— People with visual impairments struggle or require assistance from others to perform day-to-day activities. Hence, people with visual impairments suffer from a lack of independence and quality of life. For this purpose, the app proposed in the paper is a unique, creative, innovative mobile application that will aid people with visual impairments in becoming independent and improving their quality of life. This paper introduces the novel mobile application's three key features: QR (Quick Response) code scanning, OCR (Object Character Recognition), and an Indoor Object Navigation System. The QR code scanning feature allows users to make informed decisions quickly by providing comprehensive information about products or their environment. The OCR functionality reads out the printed text aloud and helps the users understand written content with the help of audio. The indoor object finder or navigation system is a noteworthy advancement in Assistive Technology. The system allows users to select a target object and then move their camera around their surroundings; once the object is detected, the app provides voice instructions about the directions of the target object to guide users towards it. By integrating these complex technologies, the novel mobile application provides an innovative approach to some of the issues encountered by those with visual impairments. The application aims to increase user independence, improve functionality, and build confidence to improve their daily experiences and engagements with the world around them.**

Keywords—Novel Mobile Application, Visually Impaired, Quick Response (QR) code, Optical Character Recognition (OCR), Indoor Navigation.

I. INTRODUCTION

People with low vision face many challenges that affect their independence and ability to carry out daily activities. These challenges ranging from finding ways in the unknown environment to accessing printed information, often make it difficult to participate fully in everyday activities. Strides have been made to provide ways to solve these problems through advanced technologies for visually impaired people. Our proposed idea is to make an app which is simple and easy to use which can solve various problems for visually impaired people by imparting features like barcode and QR scanner, OCR detector with speech transformation, color detector, path navigation, object finder, etc. Also, we will provide a guided tour in our app for new users as well as personalisation and customisation option will be there for users so that they can hide whatever app features they don't want to use. Over the time, we will take user feedback for continuous refinement of the app such as improving user interaction, achieving accurate performance and integrating new emerging technologies which can fulfill the evolving needs of this community. Our aim is to make a product which will be dynamic and evolving.

In this paper we have presented an android application which has three features that will help these low vision people that includes QR scanner, OCR-to-speech and object finder. QR is used in almost everywhere in this modern world. Using our app these QRs can be scanned by these people and they can get product information in bold fonts. Let say the users are not able to see the product information and hence not able to read it properly then they can use our next feature which is OCR-to-speech. In this first users have to take screenshot of the whatever information they are trying to view in browser or they can take pictures of the unreadable parts of the product and upload the image in our application which will perform OCR and will detect text. After that the text will read by the OCR-to-speech voice module. If users wanted to locate an object in their surroundings and move towards it to pick them then here our third featured named object finder will help them. The user has to type the object they wanted to search, then camera will get open and if any object found our voice module inform the user that current camera frame contains the searched object in it and after that it will give voice guided instructions to the user for reaching towards the object.

Several studies are reviewed in this paper to offer a complete evaluation of modern-day solutions and spotlight the

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sizable contributions and future guidelines in assistive technology. The study [1] presents an effective deep learning-based approach. The system uses a YOLOv3 neural network, which was trained on the COCO dataset and refined using both natural state and optimized QR code images. In [2], a advanced system with four subsystems is introduced: shopping list application, route optimization algorithm, store navigation subsystem, and product fetching subsystem using computer vision, the system uses QR code and barcodes, which serve as markers for navigation in a retail store layout. In paper [3] to detect text from image the authors uses an algorithm known as EAST(Efficient and Accurate Scene Text Detector) which is made using FCN with PVANet and converting it into speech they use the python package known as pyttsx3 which uses various text-to-speech algorithms.

audio design and provides those with poor visibility with useful two-dimensional spatial information for indoor navigation. In [7], an Android application for visually impaired and blind users is introduced. By providing voice commands and audio prompts, the program simplifies tasks by increasing user autonomy and accessibility for mostly text reading. In [8] study presents a smartphone app that assists individuals with partially and fully visual impairments. This application enable options for these kind of people to call, send and receive messages with more options like navigation, battery monitoring via voice commands . Intelligent and smart mobile applications for the visually impaired are described in [9, 10]. The device integrates a number of sensors with advanced algorithms to enable safe and effective communication with the environment and enable real-time

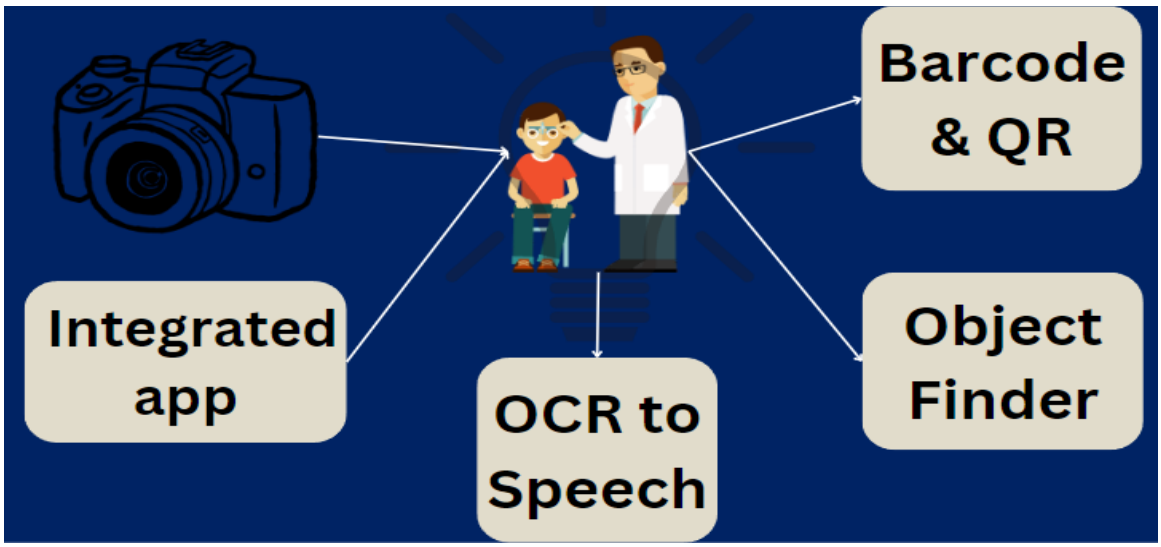

Fig. 1. Block Diagram

The document [4] provides a detailed description of the "Blind Assist" mobile application. It is intended for older and visually impaired users, and its numerous features make it easier to use and more accessible. The software uses voice assistants to read and send SMS messages to protect users' privacy. The Room Database allows for diary functionality, and entries can be stored locally to improve data security. Real-time optical character recognition (OCR) using Google Mobile Services with Text Recognizer makes printed text accessible. Via the Java Mail API, voice instructions can be used to perform email functions, which lessens the need for keyboard keys. An android application named " Eye Assistant" is discussed in [5] which can detect real time object and text by scanning them which helps visually challenged persons to know about their surroundings. This app can run on a low-end device and does not require any remote server .

II. Literature Review

The authors use the TTS method with computer vision techniques in their research [6], most notably the YOLO (You Only Look Once) algorithm, which provides efficient and fast object recognition using Convolutional Neural Networks. Furthermore, it eliminates the need for pricey RGB-D cameras and guarantees portability, at a reasonable price using the head-related transfer function (HRTF) to generate two sounds using monocular depth estimation techniques used to estimate depth from individual images, this system integrates spatial

support and feedback to users. Surveys has highlighted how these devices have the potential to improve the lives of the blind or partially sighted.

An Android smartphone crowdsourcing application for the blind and visually impaired is presented in the study [11]. The program leverages community efforts to provide real-time information and support to users, improving their own ability to perform daily tasks. A particular example of visual complexity recognition in an Android mobile application is shown in [12]. The application recognizes elements in the user's surroundings using machine learning techniques and offers advice via auditory feedback. In [13, 14], the creation and usage of a smart apps for those with vision impairments are explained using of present-day smartphones, the software offers a wide variety of capabilities, like object detection, text-to-speech conversion, and navigation assist, to offer an entire assistive answer.[15] describes the introduction of a smart contact cellphone assistance gadget for blind people to improve smartphone usability for the visually impaired. This technology includes a wide range of functions including voice commands and text-to-speech conversion. [16] introduces a system with both hardware and software which helps cellphones to identify places with obstacles in the outdoor environment and gives the route to close places.

An app named "Voisee Communicator" is described in [17] and is intended for use in hearing-blind communication

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to help users communicate. In [18], a new app named "Vocal Vision" is introduced which fulfils various inabilities of visually impaired and blind people like sending and reading emails, pathfinding or outdoor navigation, and reading SMS. The model uses various algorithms and machine learning techniques to accurately and reliably do the required tasks. [19] describes the development of a novel speech-to-text module which helps these low vision people with hearing aid to make them listen, read and understand efficiently English language. In work [20] an innovative method for assisting visually impaired people using wearable stereo vision technology is presented. This study is important because it addresses the challenges that visually impaired people face when trying to navigate their environment on their own.

III. Materials and Methods

A. Materials

- *Kotlin* is the main programming language used in the NOVEL MOBILE APPLICATION, which guarantees efficient and readable code development.
- *Android Studio*, the official IDE for Android app development, streamlines the process with tools for designing, coding, testing, and debugging, enhancing developer productivity and ensuring compatibility across Android devices.
- *Jetpack Compose*, a modern UI toolkit which simplifies the creation of advanced and user-friendly interfaces, improving development speed, code readability and maintainability.
- *Google ML Kit APIs* provides pre-trained models for integrating machine learning functionalities enabling advanced features like image recognition and text detection.
- *Computer Vision* allows the app to interpret visual information in real-time, crucial for indoor object navigation by analyzing video frames to guide users. *Machine Learning* forms the intelligent core of NOVEL MOBILE APPLICATION, enhancing the app's ability to recognize patterns and adapt to user interactions, providing a dynamic and personalized experience that evolves to meet the needs of visually impaired users. This comprehensive technology stack highlights NOVEL MOBILE APPLICATION's commitment to leveraging cutting-edge solutions for accessibility and user interaction.

B. Methods

To help visually impaired or low visioned people our app contains the following features:

1) Barcode and QR Scanning

Black lines in combination with white spaces is used create a barcode (shown in Fig. 2). A light beam is thrown upon the barcode by a barcode reader. As we know that, white color reflects light while black color absorbs it. Similar to this, the white color will reflect the light back and the dark color will absorb it when the scanner shines light on the barcode. The reflected light is detected by the sensor in the barcode reader, which then transforms it into an analog electrical signal. The

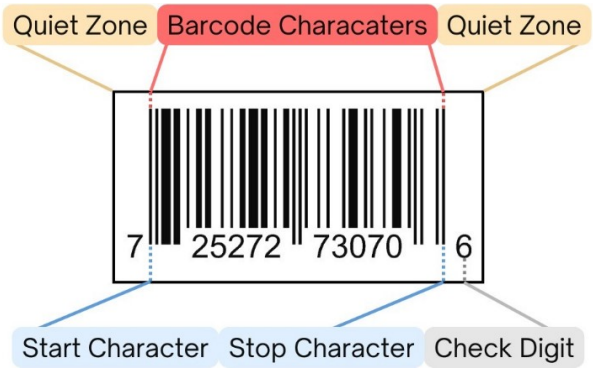
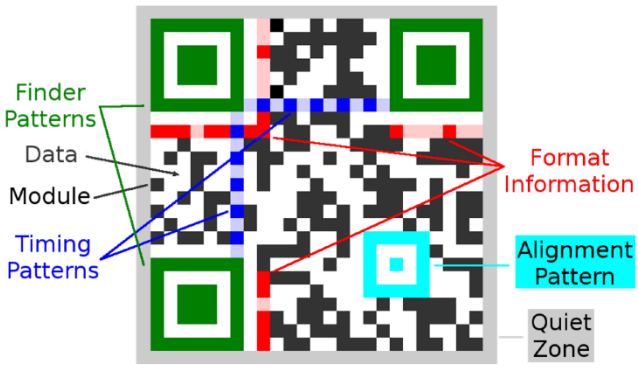


Fig. 2. Parts of a Barcode Reader

barcode reader's analog-to-digital converter transforms analog signals into digital signals. This is where next when decoder comes into play. Digital signals are then transformed into data that can be read by humans using decoders.

All QR codes have a standard structure that makes information readable(shown in Fig. 3). Let's break it down:

- Quiet Zone: The empty white-bordered (here grey) space around the QR code.
- Finder patterns: The three black squares (here green) in the top-left, top-right and bottom-left vertices of the QR code.
- Alignment pattern: A small square near the bottom-right vertex, which ensures the QR code can be read when scanned from any angle.
- Timing pattern: An L-shaped line that helps to identify individual squares within the whole code, making it possible for a damaged QR code to be read.
- Version information: Identifies which version of the QR code is being read.
- Data cells: The rest of the part of QR code which contains the actual information such as the URL, phone number, or other data.



- Fig. 3. Parts of a QR Scanner [21]

2) OCR

Step 1: Checking the Type of Document & Pre-Processing of Image

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The initial step in the OCR workflow is crucial for determining the document type and selecting the appropriate processing pipeline. Each document type, whether it be PDF, JPEG, or TIFF, has unique characteristics and content structures that require specific handling.

- Document Type Identification:

  - Documents: PDFs may contain a text layer, which can be directly extracted if available. If the text layer is absent, the document is treated as an image, requiring full OCR processing.

  - Image Files (JPEG, TIFF, PNG): These formats are treated as images without any inherent text layers, necessitating complete text extraction through OCR.

  - Scanned Documents: Often stored in TIFF format, these require thorough preprocessing to ensure text clarity for effective OCR.

- Pipeline Selection:

  - Once the document type is identified, the OCR software selects the corresponding predefined pipeline. These pipelines are developed to handle specific document formats and their associated challenges.

- Image Pre-Processing:

  - Noise Removal: This involves techniques like Gaussian blurring or median filtering to reduce image noise, which can interfere with text recognition.

  - Contrast Enhancement: By converting the document to a high-contrast black and white image, the distinction between text and background is maximized.

  - Thresholding: Adaptive thresholding methods are employed to identify and enhance light and dark areas, making the characters stand out more distinctly against the background.

  - Skew Correction: Documents may be scanned at an angle, so deskewing algorithms are applied to straighten the text lines, improving OCR accuracy.

Step 2: Character Recognition

Character recognition is the main and most important step of the OCR process, where the actual text detected from the input. This step involves novel algorithms and machine learning techniques to accurately identify and interpret characters.

- Feature Detection:

  - Algorithm Application: Feature detection algorithms analyze the shapes and structures of individual characters. These algorithms identify distinct features like edges, lines, and curves.

  - Character Segmentation: The image is divided into segments, each containing a single character. This segmentation is crucial for accurate character identification.

- Pattern Recognition

  - Stored Samples Matching: The system compares the segmented characters against a database of stored character samples. This method works well for known fonts and clear text but struggles with novel or distorted fonts.

  - Feature-Based Recognition: For unknown or distorted fonts, the system relies on the rules of character formation, such as recognizing slanted lines or curves that form part of letters.

  - Machine Learning Models: Modern OCR systems often incorporate neural networks and other machine learning models trained on vast datasets. These models enhance the ability to recognize characters in various fonts and styles, improving accuracy even with complex or degraded text.

Step 3: Post-Processing

Post-processing is essential to refine the raw output from the character recognition step, ensuring the final text is accurate and usable.

- Noise Filtering:
  - Error Detection: The initial OCR output contains noise and potential errors. Statistical analysis helps identify common OCR mistakes, such as misinterpreting 'O' as '0' or 'I' as '1'.
  - Correction Algorithms: The system uses predefined rules and dictionaries to correct these errors. For instance, it can cross-reference recognized words with a language dictionary to replace unlikely sequences with more plausible alternatives.
- Entity Combination:
  - Contextual Analysis: Recognized characters are combined into words and sentences. Contextual understanding helps the system identify and correct words based on their placement and usage within a sentence.
  - Semantic Matching: This involves linking recognized entities with their intended meanings, enhancing the overall coherence of the output text.
- Output Refinement:
  - Statistical Corrections: Leveraging big data, the system applies statistical corrections to improve text accuracy. Common patterns of errors are identified and rectified using historical data.
  - Human Review (Optional): For critical applications, human review may be incorporated to verify and correct the final output, ensuring the highest level of accuracy.

Through these detailed steps, the OCR system transforms images of text into accurate, machine-readable data, enabling a wide range of applications from document digitization to automated data entry.

3) Indoor Object Finder or Navigator

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A multi-layered model for classifying given objects into one of the defined classes is developed using CNN. Higher resolution feature maps are used to detect these objects are made possible by recent advancements and developments in deep learning with image processing. The output bounding box and object class are calculated from the input image by the object detection model MobileNet SSD. This Single Shot Detector (SSD) uses MobileNet as a backbone and can achieve fast object detection which can result in optimized performance for mobile devices.

- MobileNet Architecture

MobileNet, developed to be efficient and lightweight, is particularly suitable for mobile and embedded vision applications. It employs depthwise separable convolutions, which decompose a standard convolution into two simpler operations: depthwise convolution and pointwise convolution. This decomposition reduces computational load and model size while maintaining performance, making MobileNet an ideal backbone for object detection models like SSD. The mathematical model of MobileNet used is inspired from [23].

Standard Convolution Layer

We know that a standard convolution layer (in Fig. 4(a)) takes input feature map F with size $D_F \times D_F \times M$ where D_F is spatial width and height of feature map assumed to be square and M is the number of input channels, produces a $D_F \times D_F \times N$ output feature map G where N is the number of output channel. The output feature map assuming stride one and padding can be calculated as:

$$G_{k,l,n} = \sum_{i,j,m} K_{i,j,m,n} \cdot F_{k+i-1,l+j-1,m}$$

K is the convolution kernel (assumed to be square) of size having size $D_K \times D_K \times M \times N$ where D_K is the spatial dimension of the kernel. This standard convolution will have a computational cost of $D_K \times D_K \times M \times D_F \times D_F$. It is seen that in general the computational cost is high containing too many parameters making it unsuitable for mobile and embedded vision applications. To make it efficient for such applications we can use depthwise separable convolution.

-Deptwise Separable Convolution

This method is used to reduce computational cost and reduce the interaction between the output channels and kernel by splitting filtering and combining steps into two substeps: depthwise convolution and pointwise convolution.

Firstly for filtering purpose we use depthwise convolution (in Fig. 4(b)). Here, we apply a single filter to each input channels which can be written as:

$$\hat{G}_{k,l,m} = \sum_{i,j} \hat{K}_{i,j,m} \cdot F_{k+i-1,l+j-1,m}$$

$\hat{K}$ is the depthwise convolution kernel (assumed to be square) having size $D_K \times D_K \times M$ where D_K is the spatial dimension of the kernel and M is the number of input channels in which m^{th} filter in $\hat{K}$ is applied to m^{th} channel in F to get m^{th} channel of the filtered out feature map $\hat{G}$ which will have a computational cost of $D_K \times D_K \times M \times D_F \times D_F$.

Next to generate new features, we need an additional layer which computes the linear combination of the output of depthwise convolution using 1×1 convolution which is known as pointwise convolution (in Fig. 4(c)) found in [24] whose computational cost is $N \times M \times D_F \times D_F$.

Total computational cost of depthwise separable convolution is expressed as

$$D_K \times D_K \times M \times D_F \times D_F + N \times M \times D_F \times D_F$$

Comparing this computational cost with the standard convolution layer's computational cost we get a reduction in computation as:

$$\frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F} = \frac{1}{N} + \frac{1}{D_K^2}$$

$$\frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F} = \frac{1}{N} + \frac{1}{D_K^2}$$

MobileNet will use 3×3 depthwise separable convolution whose computation is almost 8 to 9 times lesser than standard convolution layer's which is possible as number of parameters as well as number of multiplications and additions of parameters is being reduced. We have observed that as number of parameters reduces the accuracy of image detection and computational cost reduces. So, to make a model perform better we have to maintain balance between accuracy and number of parameters used based on required applications. Moreover, it uses batchnorm and ReLU after each convolution layer in depthwise separable convolution.

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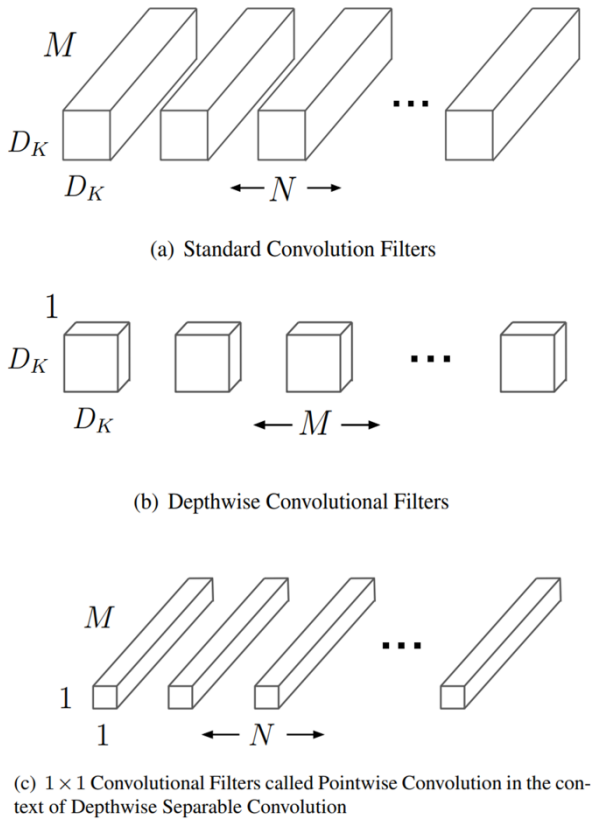

Fig. 4. Convolutions used in MobileNet as in [23]

enabling efficient and accurate object detection. (shown in Fig. 5)

- Working Algorithm

(i) Input Image: The input image is fed into the MobileNet SSD model.

(ii) Feature Map Extraction: The MobileNet backbone processes the image, generating feature maps at various levels.

(iii) Default Box Assignment: Default boxes are placed on each feature map cell, providing a grid of potential object locations.

(iv) Detection Predictions: The model computes class scores and bounding box adjustments for each default box.

(v) Non-Maximum Suppression: Redundant boxes are filtered out to keep the most accurate predictions.

(vi) Result: The remaining boxes, class labels, and confidence scores are presented as the final detection results

IV. Results and Discussions

The NOVEL MOBILE APPLICATION app's interface (shown in Fig. 6) design has proven effective in offering a robust user-friendly, reliable and easy-to-navigate experience

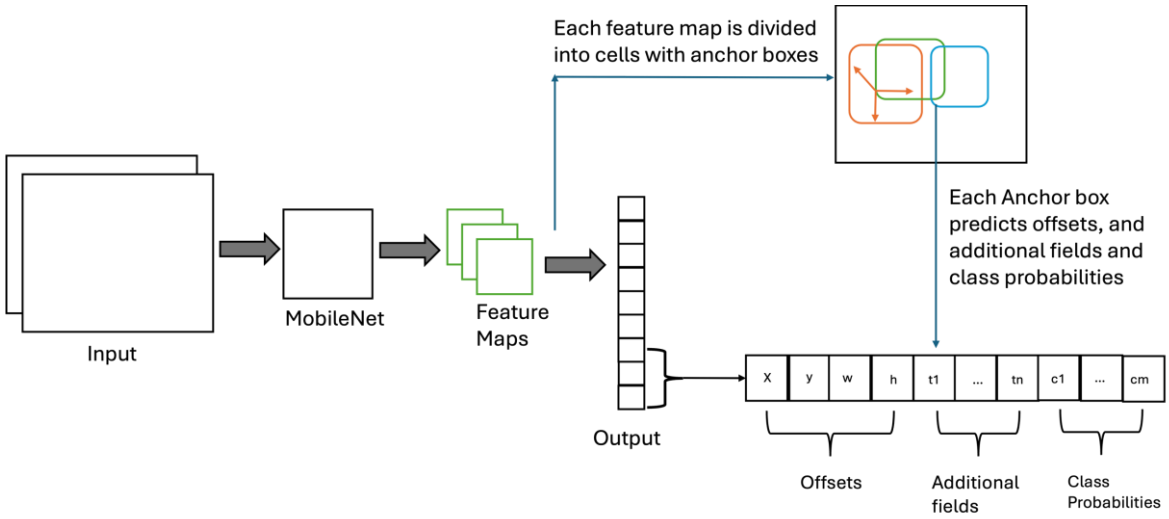

Fig. 5. Working of Object Finder / Navigator using SSD-MobileNet as in [22]

- Single Shot Detector (SSD)

SSD reframes the object detection problem as a single regression problem. It predicts bounding boxes and class scores for objects directly from feature maps with-out the need for a separate region proposal stage, as used in two-stage detectors like Faster R-CNN. SSD utilizes a set of default (prior) boxes of different aspect ratios and scales at each feature map location,

for people with visual impairments. The app ensures accessibility and simplicity through straightforward navigation, easy to use interface and basic command options. The positive outcomes highlight NOVEL MOBILE APPLICATION's commitment to providing a seamless and empowering digital tool for the visually impaired community, aligning with its mission to enhance independence and accessibility in daily tasks.

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To develop the mobile application we have use the following key components:

- Programming Language and Development Environment

  Kotlin is the primary programming language used and found better than Java due to its more efficiency, readability and easy to use,

  Android Studio the official IDE (Integrated Development Environment) for development of android applications is utilized which has offered various tools for designing, coding and debugging of the application.

- User Interface Development

  Jetpack Compose is used for UI interface development which is a modern toolkit via Google. It makes the creation environment advanced and has responsive UIs which makes the coding less, improving development speed and maintainability.

- Machine Learning Integration

  We have integrated Google ML kit APIs to provide our app advanced machine learning capabilities to our app. It contains pre-trained models for QR code scanning and OCR purposes. Also, whatever text is being detected we have integrated a voice command which will read it and user can select that text and copy it too. We have also integrated and used SSD-MobileNet model for object finder or navigator feature in our application (discussed in section III).

- Computer Vision Capabilities

  Computer Vision plays a very crucial role in our application by enabling and analysing visual information. All the three features in our app requires camera access this technology helps to process the images as well as video frames from the camera in real-time. It allows the app to provide real-time feedback to assist with navigation and object identification.

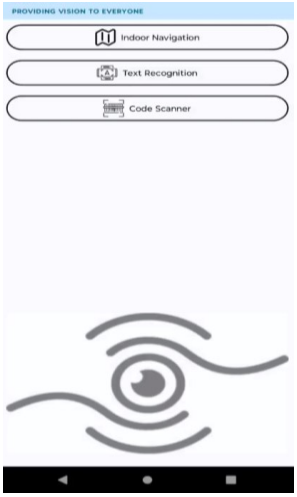

Fig. 6. Novel mobile app interface

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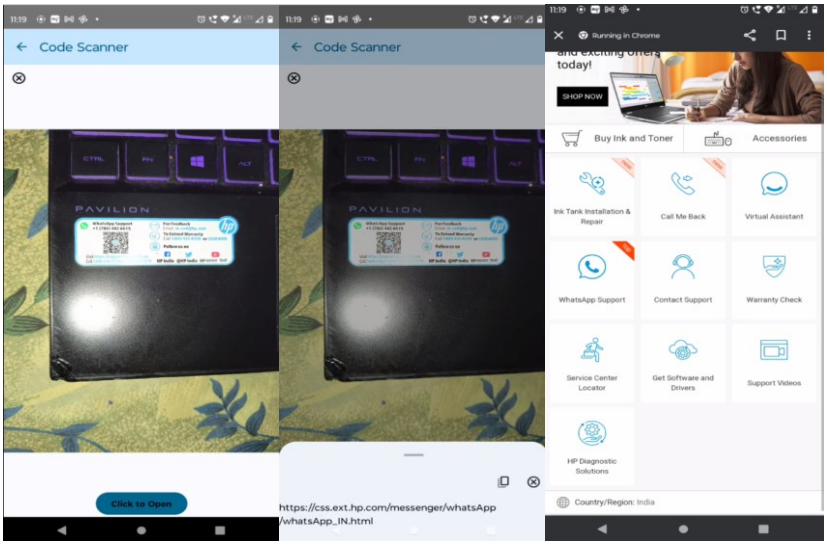

Fig. 7. Scanned QR using app and opening the scanned website

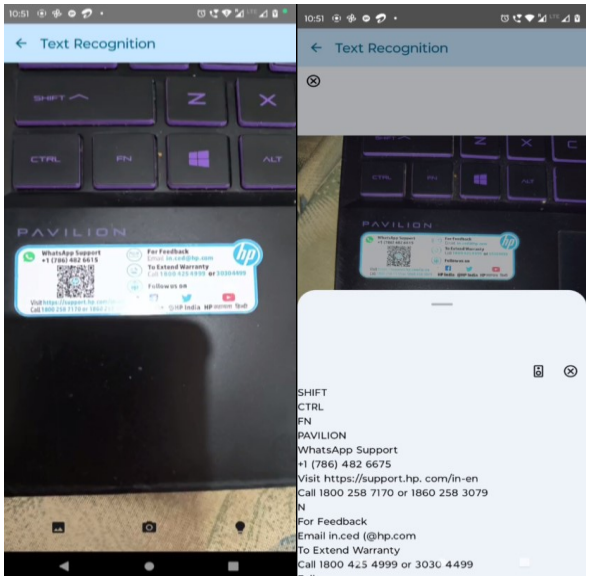

Fig. 8. Feed an image to OCR feature of app and its output

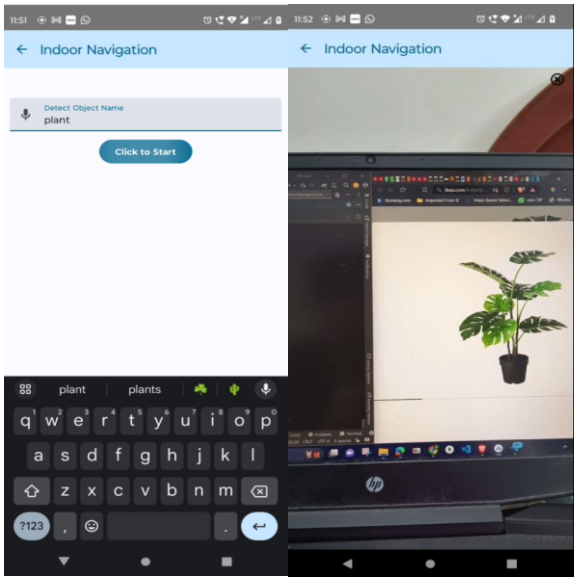

Fig. 9. Indoor navigation feature to find plant in image and get directions to move towards it

The barcode and QR code functionality within the NOVEL MOBILE app has shown exceptional efficiency in detection and information extraction in Fig.7. Through advanced algorithms, the app swiftly identifies QR codes, allowing users with visual impairments to effortlessly gather information embedded in these codes. Particularly noteworthy is the app's capability to seamlessly handle QR codes that may contain links to websites. Once detected, the app initiates a redirection process, efficiently navigating users to the specified website. This seamless integration of QR code scanning not only enhances accessibility by providing instant access to textual information but also empowers users by enabling swift exploration of online resources, contributing significantly to the overall effectiveness and user satisfaction of the app.

The implementation of Optical Character Recognition (OCR) in the NOVEL MOBILE app has demonstrated exceptional efficiency and accuracy in swiftly detecting and extracting text from input camera images which is being shown in Fig.7. This advanced functionality ensures users with visual impairments can seamlessly access textual information from diverse sources in real-time. The app's user-friendly design highlighted by the easily accessible red-circled button, allows users to effortlessly convert the detected text into speech, providing immediate and accessible content. The integration of OCR and Text-to-Speech technologies not only enhances accessibility but also reflects in our apps'

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commitment to fostering independence and inclusivity for individuals with visual impairments.

The object finder/navigation feature in the NOVEL MOBILE app, powered by SSD-MobileNet, has also resulted in an exceptional performance in real-time object detection and spatial awareness. Leveraging dynamic feature maps, the app provides users with a complete understanding of their environment, allowing them to identify and locate objects seamlessly. The audible directional guidance adds an innovative dimension, offering users a real-time spoken feedback on object locations relative to the camera.

V. Conclusion and Future Scope

A. Conclusion

In conclusion, the development of NOVEL MOBILE BASED - an app designed for empowering visually challenged individuals marks a significant stride in addressing the difficulties and problems faced by this vision impaired community. The combination of features such as a QR scanner, OCR recognition with text-to-speech capabilities, and innovative indoor object navigation using continuous video frames is poised to enhance accessibility and independence for visually impaired users. The app's ability to seamlessly recognize and convey information about products and textual content, coupled with its real-time indoor navigation, reflects a commitment to overcoming barriers and fostering a more inclusive digital environment for these people.

B. Future Scope

Looking ahead, NOVEL MOBILE APPLICATION is expansive and promising. Continuous refinement of the app's algorithms and integration with evolving technologies could lead to enhanced accuracy and speed in recognizing a broader range of objects and text formats. The expansion of language support and incorporation of more intuitive voice commands could further enhance user experience. Additionally, collaborations with institutions and organizations could facilitate the integration of geospatial data for outdoor navigation, ensuring a com-prehensive solution for users in diverse environments. The ongoing commitment to re-search and development will position NOVEL MOBILE APPLICATION at the forefront of assistive technology, continually evolving to meet the dynamic needs of the visually impaired community and promoting a more inclusive and accessible world.

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